Key: Which Rung?

84-355 Democracy's Data | Levels of aggregation | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure. The brief (`levels-of-aggregation-brief.Rmd`) and the lab (`levels-of-aggregation.Rmd`) run the *same numbered steps* — the brief in prose with figures, the lab with the code exposed. **Students should read the brief first.** Step numbers below refer to both.

Where it goes in the term. After `census-geography` (which units nest) and `precinct-geography` (what happens when a unit moves). Those two are about the shapes; this one is about **which shape to use**, and it is the one that generalises beyond elections. It can be run any time after Session 14, and it sets up `rpv` so completely that running it the session before turns the RPV lab into a confirmation rather than a surprise.

It reuses seven other labs' data and downloads nothing. If a student asks where the files came from, the answer is "from work you have already done," which is worth saying out loud.

Every number below was run against the committed files.

The session in one sentence

Election data comes on a ladder from the ballot to the nation; every step up buys coverage and comparability, permanently destroys the ability to link things that co-occurred below, and creates a new opportunity to be confidently and impossibly wrong.

The findings

```r
f1 <- function(x, k) formatC(x, format = "f", digits = k, big.mark = ",")
data.frame(
  finding = c("ballots in the Alaska cast vote record",
              "... as a share of all 2024 ballots (%)",
              "ranked Begich first and Peltola second",
              "Georgia two-party Dem share, the state total (%)",
              "precinct-level variation surviving at county level (%)",
              "Goodman's answer for Georgia's mail vote (%)",
              "the truth Georgia published (%)",
```

```
             "Goodman at county level (%)"),
  value = c(f1(cf$value[1], 0),
            f1(100 * cf$value[1] / ld$units_nationally[1], 3),
            f1(cf$value[cf$quantity == "ranked Begich first and Peltola second"], 0),
            f1(lst$mean_dem_share[lst$level == "state"], 2),
            f1(lst$variance_retained_pct[lst$level == "county"], 1),
            f1(ee$est_by_mail[1], 1), f1(TRUTH, 1), f1(ee$est_by_mail[2], 1)))
```

Finding	Value
ballots in the Alaska cast vote record	340,778
… as a share of all 2024 ballots (%)	0.215
ranked Begich first and Peltola second	6,663
Georgia two-party Dem share, the state total (%)	48.89
precinct-level variation surviving at county level (%)	69.9
Goodman's answer for Georgia's mail vote (%)	194.0
the truth Georgia published (%)	64.5
Goodman at county level (%)	233.4

- **340,778 ballots** in the only individual-level American election file in this course — **0.215%** of the national total.
- **6,663 Alaskans ranked a Republican first and a Democrat second.** Unknowable from any precinct total.
- **The vote-weighted mean survives the climb; 30.1% of the variation does not.**
- **Goodman's ecological regression returns 194.0% where the truth is 64.5%**, and **233.4%** one rung higher.
- **Real counties push the correlation to 0.760; none of 500 random regroupings of the same precincts reached it.**

The three things to get said

1. The secret ballot is the reason the problem exists. Students arrive assuming individual-level data is missing because nobody bothered to collect it. It is missing **on purpose**, and the purpose is good. Say this before Step 6, so that when the method fails they read it as the price of an institution rather than as a technical embarrassment.

2. Aggregation is not zooming out. The photograph analogy is the one they have. Break it: a photograph you zoom out of can be zoomed back in. Once you add 6,663 Begich-then-Peltola ballots into a precinct total, **no amount of cleverness gets them back.** The information is not hidden; it is gone.

3. The answer of 194% is not a bug and not a fluke. It is what a between-unit relationship looks like when you read it as a within-unit one. Precincts that mailed heavily were Democratic precincts, and Democratic precincts were more Democratic among mail voters *and* election-day voters. The regression cannot tell those apart, so it puts the whole difference on the method of voting.

The line for the board: *the number you get depends on the unit you chose, and nothing in the data will tell you that you chose wrong.*

The move that makes this session land

Take the prediction seriously. Before Step 6, make the room write down two numbers: their guess for the Democratic share of Georgia's mail vote, and the largest value that answer could possibly take. Everyone writes 100 for the second. Then run the chunk.

The whole session turns on ninety seconds of that. They have just written down a constraint the method violates, and they wrote it themselves.

If you want a second beat: ask what they would have concluded had Georgia *not* published the crosstab. The honest answer is "we would have believed it, adjusted to something under 100, and published."

Walking the steps

Steps	What they do	Min	Watch for
1–3	The ladder; the Alaska ballots; what one rung up costs	10	The Begich-then-Peltola number. It only exists at the bottom.
4	One election at four levels; the mean holds, the spread collapses	8	48.90% at precinct, county and state. Aggregation does exactly what it promises.
5	The residuals; Alaska has no counties; the district rung	8	A rung is not a rung everywhere. 44 rows of the county file are not counties.
6	**The pivot.** Truth 64.5%; Goodman 194.0%	14	Take the prediction first. The two lines share a cloud and differ only in slope.
7	County level: 233.4%. The random-regrouping null	10	Coarser is worse, and the lift comes from *these* lines, not from coarseness.
8–10	The RPV case; what the aggregate is for; the decision table	12	The decision table is the deliverable. Spend real time on it.

Step 6 is the pivot, and it is stronger than the usual ecological-fallacy demonstration because there is an answer key. Robinson's 1930 example asks students to take the individual-level truth on faith. This one publishes it in the row above.

If you are short on time, cut Step 5's district material and protect Steps 6, 7 and 10. Step 10 is what they will actually use again.

Option A — Break the regression differently

Verified output

```
o <- ep[order(-ep$votes), ]
a <- data.frame(
  variant = c("unweighted", "weighted by votes", "largest 500 precincts",
              "smallest 500 precincts"),
  goodman_estimate = round(c(goodman(ep, rep(1, nrow(ep))), goodman(ep),
                             goodman(head(o, 500)), goodman(tail(o, 500))), 1))
a$distance_from_truth <- round(a$goodman_estimate - TRUTH, 1)
a
```

Variant	Goodman estimate	Distance from truth
unweighted	193.8	129.3
weighted by votes	194.0	129.5
largest 500 precincts	185.8	121.3
smallest 500 precincts	215.6	151.1

All four are impossible. The best of them, the largest 500 precincts, is still 121 points above a truth of 64.5%.

3/4: Reports the four numbers and notes that weighting barely matters.

4/4: Says *why* weighting barely matters — the bias is in the slope, not in how the units are counted, so re-weighting the same misspecification cannot rescue it. Notices that the smallest precincts are the worst (215.6%), which is the opposite of the usual instinct that finer units are safer.

4/4, best: Answers the question the option actually asks — *does the ranking tell you anything you could have known without the answer key?* **No.** Nothing in the four fits distinguishes them; the diagnostics are fine for all four. The only reason we can say which is least bad is that Georgia published the crosstab, and for the racially polarized voting question no state ever does.

Option B — Find the level where the answer flips

Verified output

```
set.seed(84355)
merge_counties <- function(k) {
  o2 <- ec[order(ec$dem_pct), ]
  cm <- rep(seq_len(k), each = ceiling(nrow(o2) / k))[seq_len(nrow(o2))]
  z <- aggregate(cbind(votes, dem, mail) ~ cm, data.frame(o2, cm), sum)
  z$mail_pct <- 100 * z$mail / z$votes; z$dem_pct <- 100 * z$dem / z$votes
  goodman(z)
}
b <- data.frame(units = c(nrow(ep), 159, 80, 40, 20, 10, 5),
                goodman_estimate = round(c(goodman(ep), sapply(
```

```r
                    c(159, 80, 40, 20, 10, 5), merge_counties)), 1))
b$still_impossible <- b$goodman_estimate > 100
b
```

Units	Goodman estimate	Still impossible
2653	194.0	TRUE
159	233.4	TRUE
80	246.3	TRUE
40	264.8	TRUE
20	303.8	TRUE
10	313.2	TRUE
5	324.8	TRUE

It never crosses 100%. The estimate gets *worse* as the units get coarser — 325% at five units — and it is above 100 at every level we can build.

3/4: Produces the series and reports that the estimate rises with coarseness.

4/4: States the answer to the question as asked: **the crossing point does not exist.** There is no unit at which this method returns a possible number, so "choose a coarser unit" and "choose a finer unit" are both wrong answers; the problem is the method's assumption, not the resolution.

4/4, best: Notices that the merging *rule* matters as much as the number of units — merging counties by similarity of Democratic share (as above) makes it worse monotonically, while merging them at random does not. **That is Openshaw's zoning effect, discovered by the student rather than read.** A student who compares two merging rules and says so has done the best available version of this option.

Option C — Pick a question, defend a rung

No verified output; grade the reasoning. The decision table is the model.

```r
dc[c(3, 6, 8), c("question", "right_rung", "the_trap")]
```

Question	Right rung	The trap
Is voting in this county racially polarized?	precinct	The estimate is an inference, not a measurement. Report the bounds.
How many mail ballots were rejected, and for what reason?	election jurisdiction	'Jurisdiction' is a township in Wisconsin and a county in Texas.
Does this map convert votes into seats fairly?	congressional district	Districts do not nest into counties; someone had to disaggregate blocks.

3/4: Names a rung and gives a reason that is about the question rather than about convenience.

4/4: Does all three things the option asks — what is lost from the rung below, what is risked at the rung above, and a number from this lab that supports the choice. The most commonly available number is the 30.1% of variation that disappears going precinct-to-county.

4/4, best: Picks a question where the right answer is *not* the finest available unit, and defends it. Ballot rejection rates are the cleanest case: the jurisdiction is the right rung not because it is convenient but because **rejection is an administrative act performed by that office**, and a precinct-level rejection rate would be measuring something that does not happen at precincts.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is treating the rung as a *choice with consequences* rather than as a property of the file they were handed.
- **A 2** is "you should use the smallest unit available." That is the instinct the lab is designed to break, and Option A's smallest-500 result is the counterexample to hand back.
- **A 2** is also "ecological inference is unreliable, so don't use it." Courts use it; there is nothing else. The 4/4 position is that it is unreliable *and* necessary, and that the honest response is to report the bounds alongside.
- **Reward anyone who reconsiders an earlier lab.** Any claim they made this term at county or precinct level is now open to the question of whether the rung was right.

Anticipated questions

- *"Then how does anyone do voting-rights cases?"* — With ecological inference, under oath, with experts on both sides. **`rpv` is that lab.** The honest summary: the bounds are correct and nearly useless, the point estimates are useful and sometimes impossible, and courts have accepted the latter since *Gingles* (1986).
- *"Why doesn't Goodman just constrain the answer to 0–100?"* — Constrained versions exist and King's method is essentially that plus a model. Constraining the output does not fix the assumption; it hides the symptom that told us the assumption failed. **The 194% is the most useful thing the method produced.**
- *"Is the 194% because of the third-party votes in the denominator?"* — No. The denominator is all presidential votes at both ends, truth and estimate alike, so the comparison is like-for-like. Two-party denominators move both numbers a little and change nothing.
- *"Couldn't we just get cast vote records everywhere?"* — Some states publish them; Texas issued emergency guidance in June 2024 telling officials to redact more, after a voter was linked to a ballot. This is a live legal fight, not an oversight.
- *"Why is Alaska missing from the county file?"* — Alaska has no counties. The file substitutes 37 state house districts and DC substitutes 7 wards. Good moment to point back at `census-geography`.
- *"Why is the mean of the 14 district shares not the state share?"* — Because the district file carries percentages and no vote counts, so that mean is unweighted. Say so; it is also the lab's own listed weakness.

Connections

- `census-geography` — which units nest. That lab is about *containers*, this one about *choosing among them*. Run that first.
- `precinct-geography` — what happens when a unit moves between elections. Supplies the trap in the "neighbourhood swing" row of the decision table.

- `cast-vote-records` — the bottom rung, in full. This lab borrows its file and points out that it was itself aggregated before being committed.
- `ga-precinct-returns` — supplies the Georgia returns and the vote-method breakdown that makes the pivot possible.
- `areal-units` — the modifiable areal unit problem on its own. Step 7's random-regrouping null is that lab's question, met in passing.
- `rpv` — the payoff. Ecological inference exists *because* the ladder starts at the precinct. Run this lab first and the RPV session needs no setup.
- `vote-dilution` — where Step 8's failure lands in a courtroom. Same county, next question.
- `data-sources` — the 11-vote and 10,874-vote disagreements in Step 5 are that lab's finding, met again one rung higher.
- `eavs` — supplies the national unit counts and the "election jurisdiction" rung, which is the one students have never heard of.